

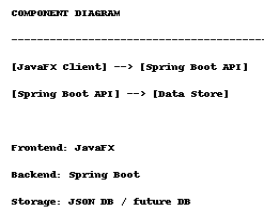
Project Report – Health & Diet Analyzer for Hostel Students

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1. Introduction

This project provides a diet logging and nutritional analysis system for hostel students using a Java-based implementation with optional Spring Boot and JavaFX modules.

2. System Architecture



3. Use Case Diagram

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USE CASE DIAGRAM
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Actor: User

Use Cases:
- Register/Login
- Log Meal
- View Dashboard
- Get Recommendation

User --> (Register/Login)
User --> (Log Meal)
User --> (View Dashboard)
User --> (Get Recommendation)
```

4. Sequence Diagram

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SEQUENCE DIAGRAM
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User -> Client: Open App
Client -> Server: POST /meals
Server -> DB: Store meal
DB -> Server: OK
Server -> Client: OK

Client -> Server: GET /summary
Server -> DB: Aggregate
DB -> Server: Totals
Server -> Client: Summary + Risk
```

5. Implementation Summary

The system includes: Java console app, Spring Boot REST API, JavaFX client, diagram set, and structured data storage.

6. Conclusion

The application demonstrates core software engineering principles including modular design, persistence, architecture, UML modelling, and multi-layered implementation.